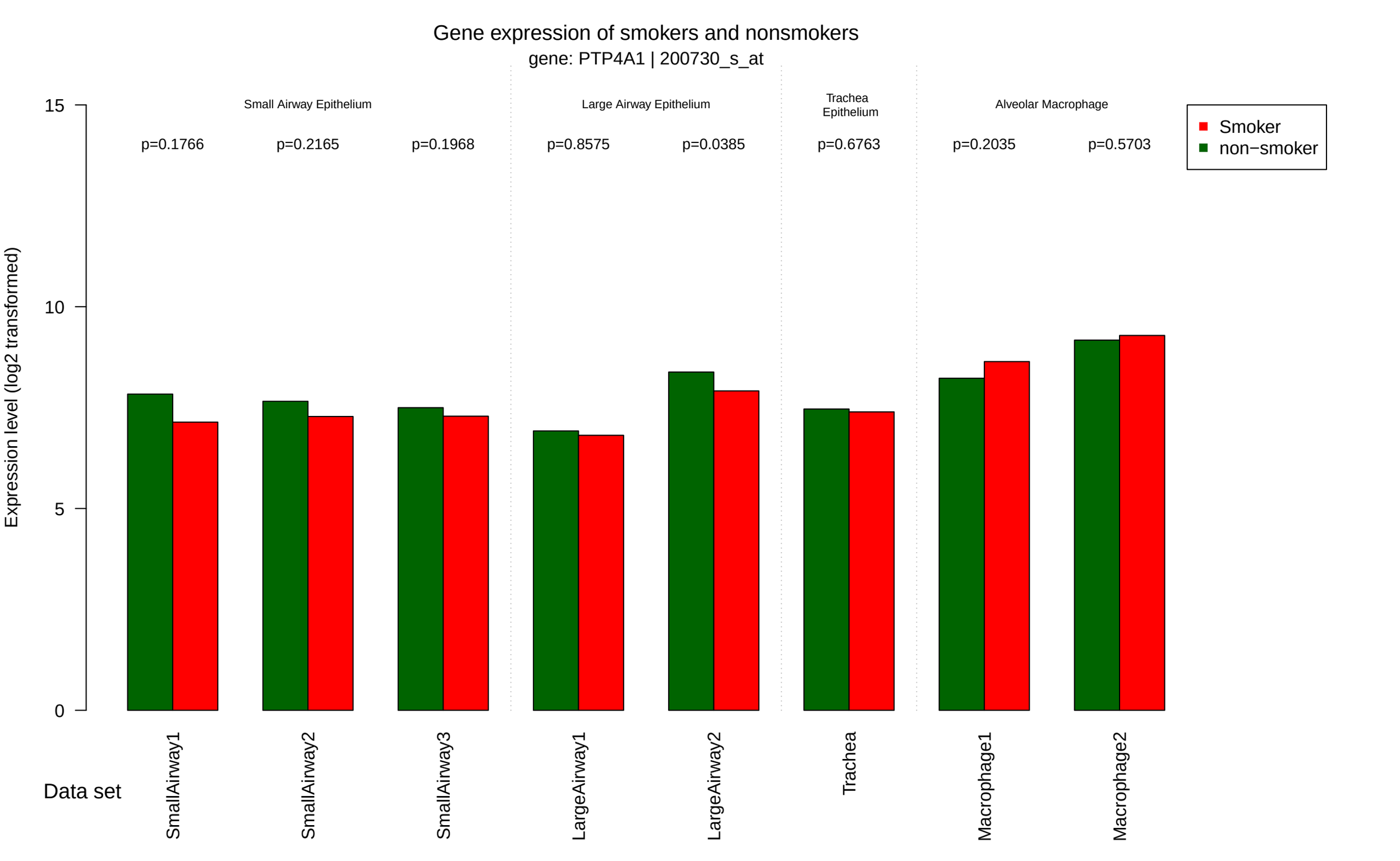
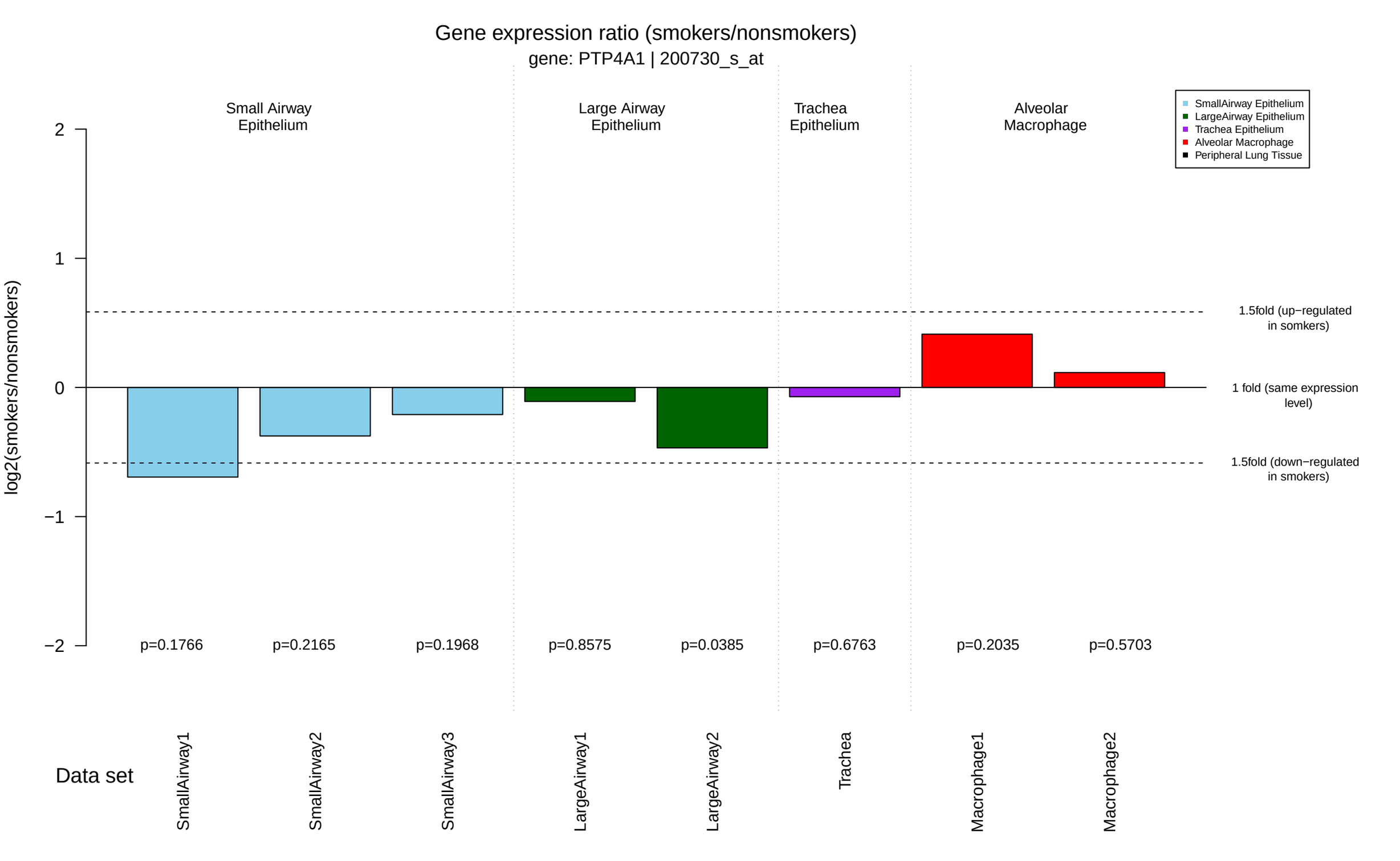
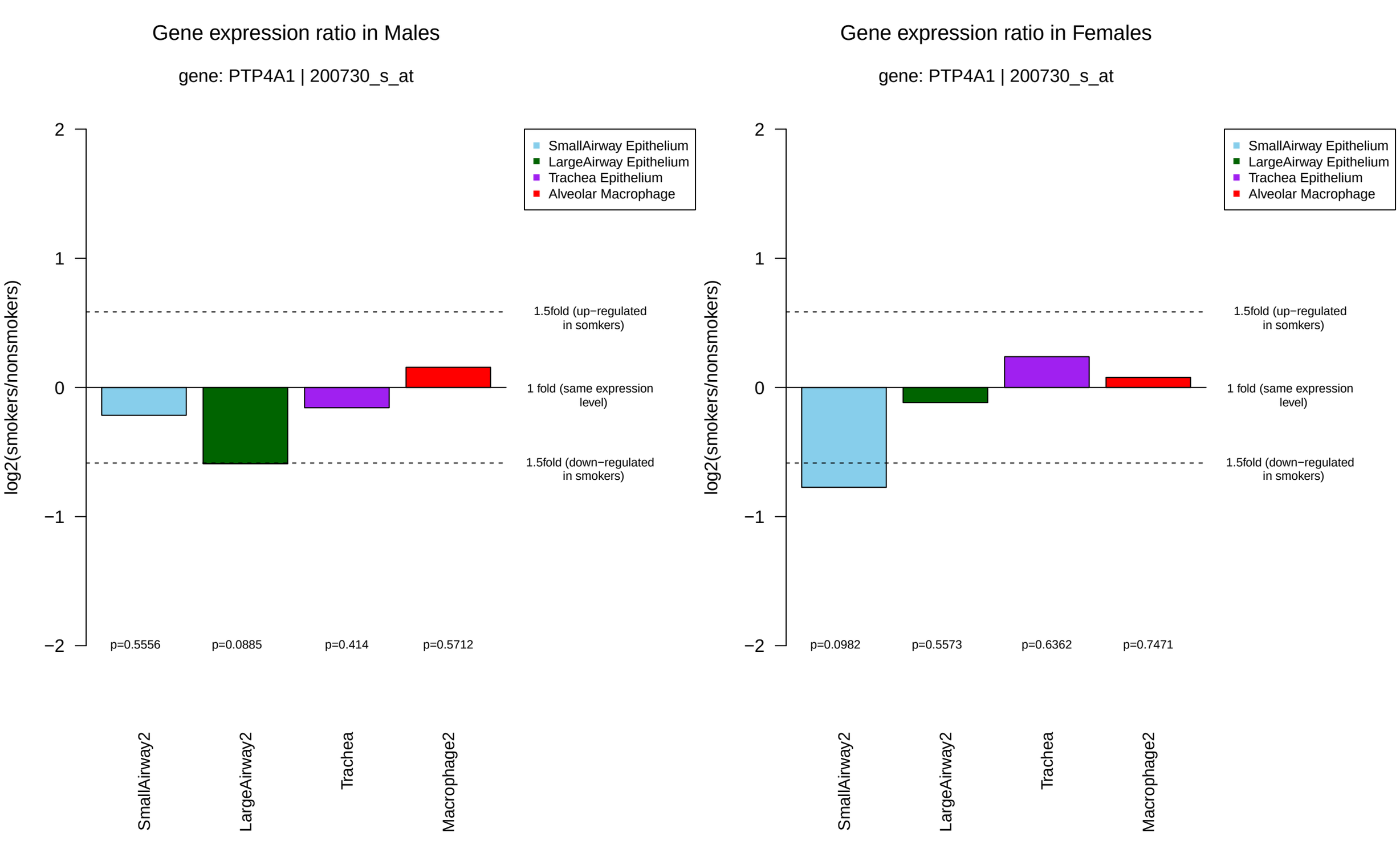
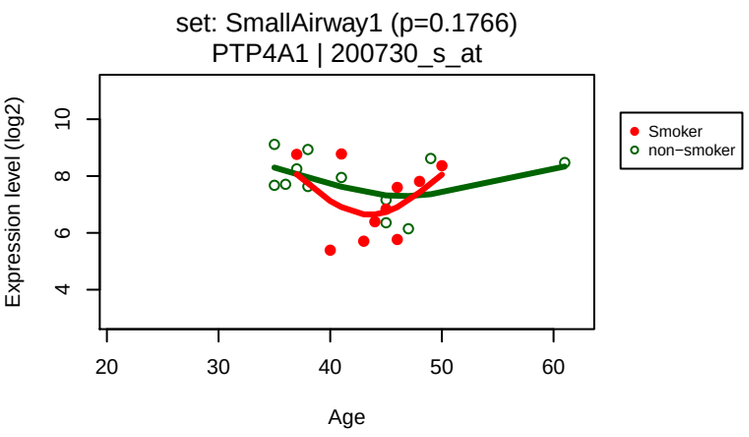
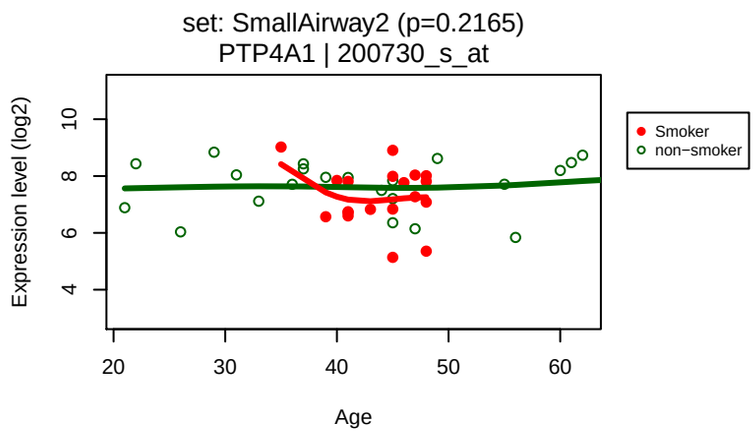
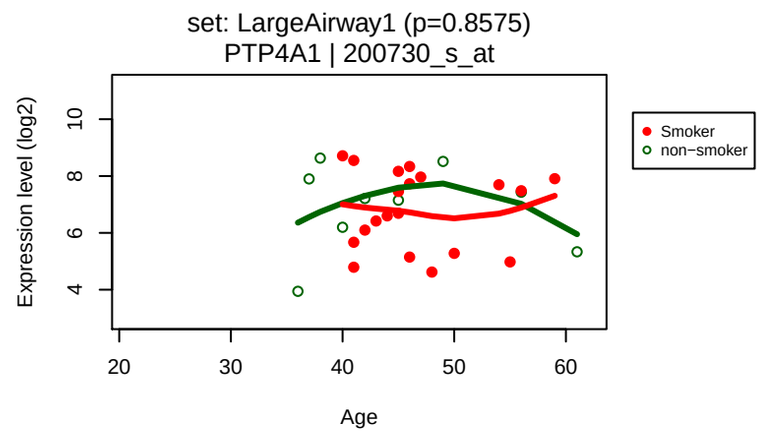

Gene expression of smokers and nonsmokers
gene: PTP4A1 | 200730_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1766
p=0.2165
p=0.1968
p=0.8575
p=0.0385
p=0.6763
p=0.2035
p=0.5703
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

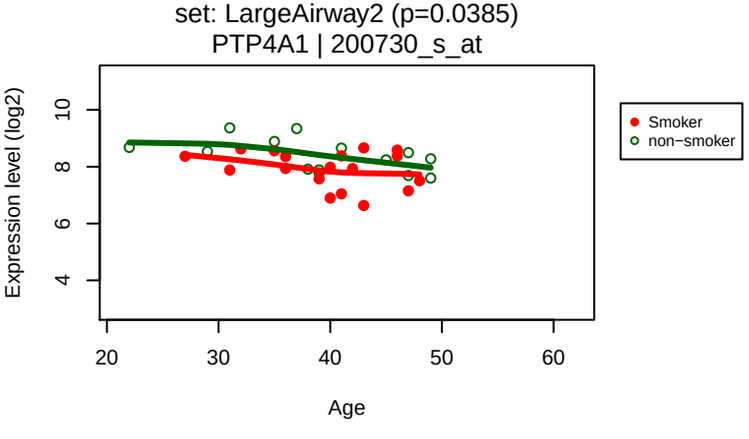

Gene expression ratio (smokers/nonsmokers)
gene: PTP4A1 | 200730_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1766
p=0.2165
p=0.1968
p=0.8575
p=0.0385
p=0.6763
p=0.2035
p=0.5703
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

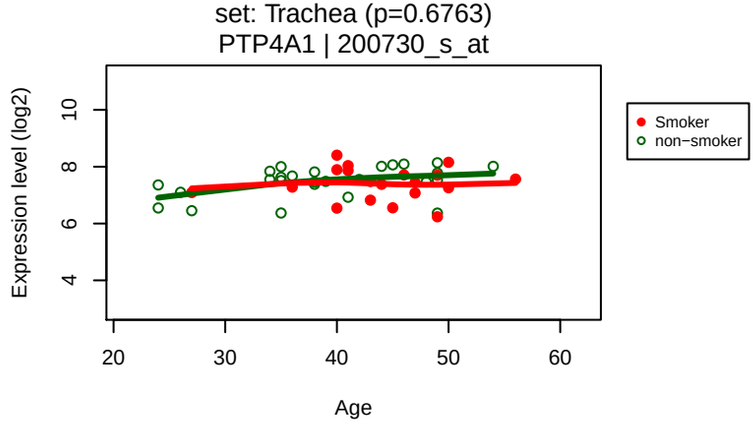

Gene expression ratio in Males
gene: PTP4A1 | 200730_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5556
p=0.0885
p=0.414
p=0.5712
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTP4A1 | 200730_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0982
p=0.5573
p=0.6362
p=0.7471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

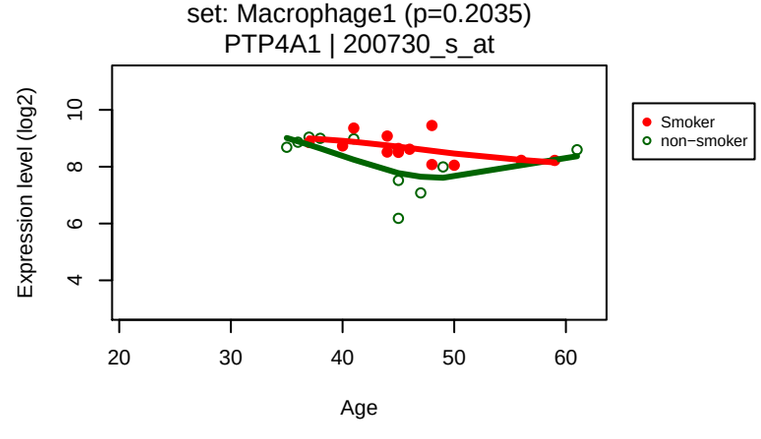

set: SmallAirway1 (p=0.1766)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker

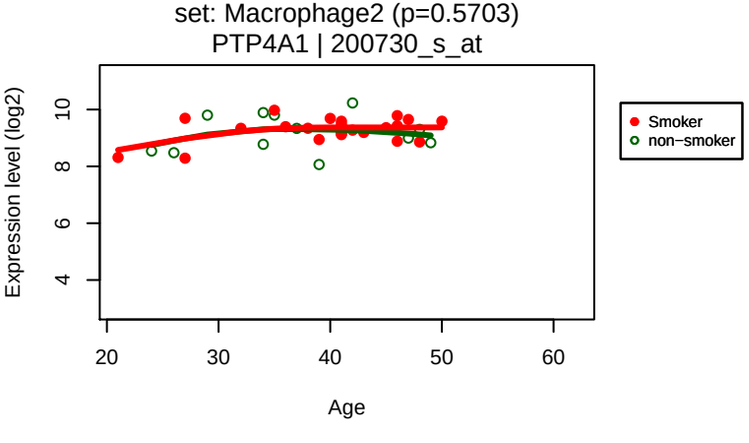

set: SmallAirway2 (p=0.2165)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker

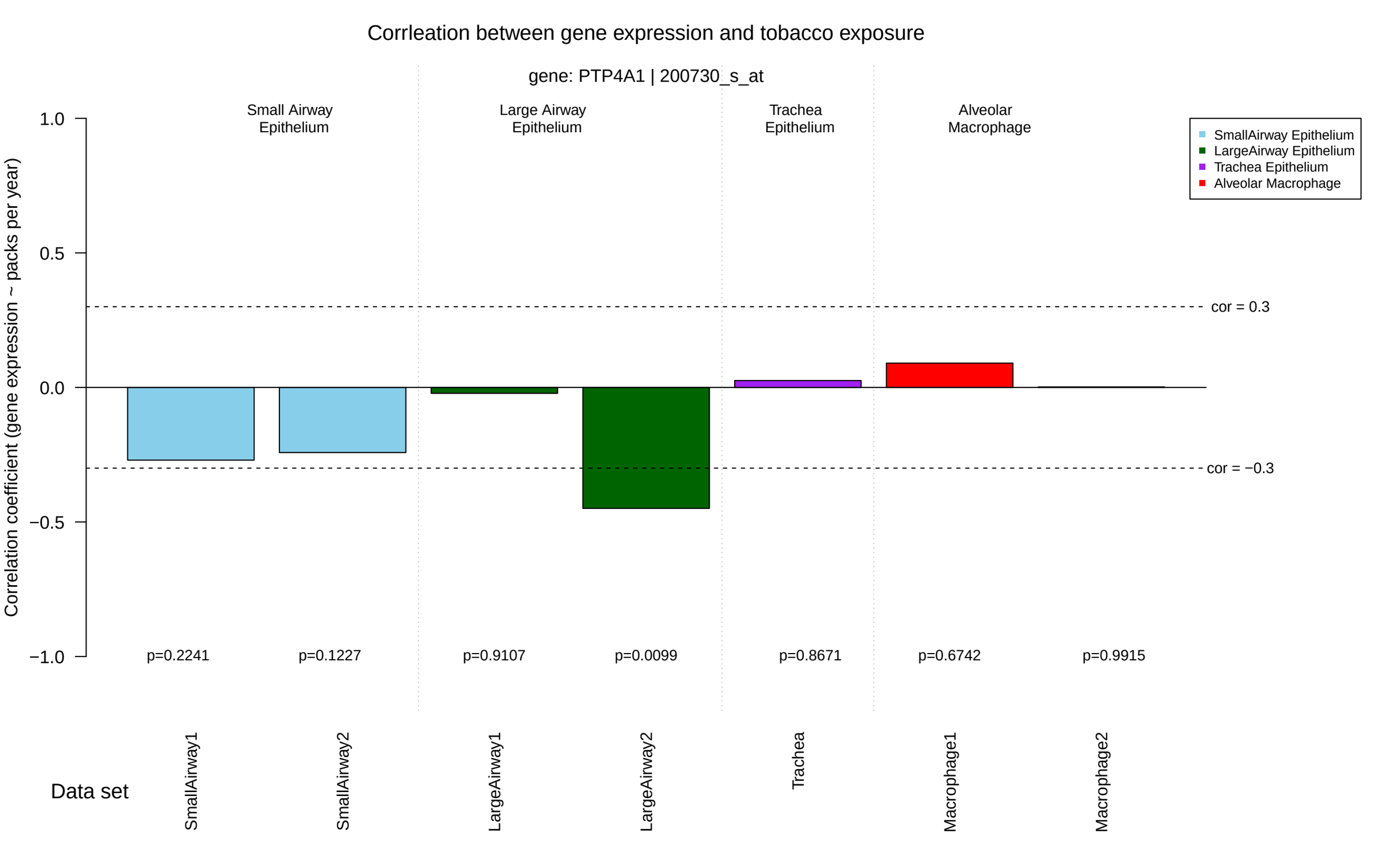

set: LargeAirway1 (p=0.8575)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0385)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6763)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2035)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5703)
PTP4A1 | 200730_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTP4A1 | 200730_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2241
p=0.1227
p=0.9107
p=0.0099
p=0.8671
p=0.6742
p=0.9915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2